In RAG We Trust? A Controlled Measurement of How Open Language Models Handle Poisoned Retrieval

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Abstract

Retrieval-augmented generation trusts the passages it retrieves. We measure how three open instruction-tuned models of 7–8 billion parameters (Qwen2.5, Llama-3.1, Mistral) answer 500 multi-hop HotpotQA questions when none to all of the answer-bearing passages are replaced by poisoned twins, and whether prompts asking the model to check the passages for consistency or reliability help, with scoring rules and confirmatory tests fixed in advance. Attack success rises from almost zero to 54.5–58.3 % as poisoning grows, overtakes accuracy once the false passages are the majority, and overrides answers the models give correctly without retrieval. Meta-prompts modestly reduce attack success in 5 of 6 comparisons, mainly through abstention, but reduce accuracy on clean evidence for Mistral. Responses that state a conflict are less often poisoned, an association rather than a defence, and the poison's framing barely matters.

Keywords: retrieval-augmented generation, knowledge conflicts, data poisoning, robustness, evaluation of language models

1 Introduction

Retrieval-augmented generation (RAG) lets a language model answer with passages that a retriever selects from documents the model was not trained on (Lewis et al. 2020). A RAG system is therefore only as reliable as what it retrieves, and it implicitly trusts the retrieved text. That trust is easy to break: Zou et al. (2025) showed that five crafted texts per target question, injected into a knowledge base of millions of

documents, make a RAG system return the attacker's answer in about 90% of cases, and a recent survey observes that most work on RAG has targeted accuracy and efficiency rather than factuality and robustness (Zhou et al. 2024).

Two empirical studies are closest to ours. Jin et al. (2024) found that models tend to follow the answer supported by the majority of retrieved passages and show confirmation bias, and Wu et al. (2024) that models adopted perturbed answers over a correct prior more than 60% of the time, less often the less realistic the error.

These studies establish that language models can be misled by retrieved text. We ask a narrower, practical question: *how reliably do open, instruction-tuned language models answer when part of the retrieved evidence is deliberately false, and can simple prompt-level instructions make them more reliable?* A user of an open model cannot retrain it but can change what it is told, so whether an instruction to check the passages protects, and at what cost, is worth measuring.

Our contribution is a measurement, not a new method. We use 500 multi-hop questions from HotpotQA (Yang et al. 2018), with contexts in which the share of false passages ranges from none to all, four kinds of poison that share one false answer and differ by a framing sentence, and three prompts that differ by one sentence; three open 7–8B models answer every condition, and a lexical retriever over a poisoned collection shows how often such contexts arise. Scoring rules and confirmatory tests were fixed before the main run and validated by hand on a pilot set. We find that all three models follow the majority of the passages, that the meta-prompts lower attack success mainly by making the models abstain, and that stating a conflict goes with resisting it but does not guarantee it.

2 Research Question and Methodology

2.1 Research questions and hypotheses

We split the main question into four research questions. **RQ1** (robustness): how do accuracy and attack success change as the share of poisoned passages grows from none to all? **RQ2** (meta-prompts): do prompts asking the model to check the passages for consistency, or to judge their reliability, lower attack success, and at what cost on clean contexts? **RQ3** (conflict awareness): how often do models state that the passages conflict, and are such responses less often poisoned? **RQ4** (secondary): do these answers differ between poison types and models?

The hypotheses were fixed before any experiment. **H1**: attack success rises with the poison ratio, most steeply where the poisoned passages stop being outnumbered (Jin et al. 2024), and is high with all passages poisoned even for questions answered correctly without passages (Wu et al. 2024). **H2** (our expectation): meta-prompts help mainly when clean and poisoned passages are mixed, little when all are poisoned, and cost some accuracy and abstentions on clean contexts. **H3** (our expectation): conflict statements peak at intermediate ratios, and flagged responses are less often poisoned. **H4**: more plausible or authoritative poisons are more effective (Wu et al. 2024); models differ, in no predicted direction. The test of H1 concerns the rise only; RQ3 is an association, since the experimenter does not control whether a model states a conflict.

2.2 Problem definition

Each question q_i ($i = 1, \ldots, N$) has a gold answer a_i^*, one false *target answer* $\tilde{a}_i \neq a_i^*$, $k = 4$ clean answer-bearing passages $c_{i1}, \ldots, c_{i4}$ (the answer passage and three paraphrases), a bridge passage b_i that is never poisoned, and, for each poison type τ, a poisoned twin $\tilde{c}_{ij}^{\tau}$ of each clean passage that supports $\tilde{a}_i$ and does not contain a_i^*. For $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$, the $m = \rho k$ poisoned slots are the first m entries of a random permutation σ_i drawn once per question:

$$C_{\rho,\tau}(q_i) = \varphi_i\big(p_{i1}, \ldots, p_{i4}, b_i\big), \qquad p_{ij} = \begin{cases} \tilde{c}_{ij}^{\tau} & \text{if } \sigma_i^{-1}(j) \leq m, \\ c_{ij} & \text{otherwise,} \end{cases} \tag{1}$$

with a passage order φ_i fixed per question. The poisoned sets are nested in ρ and shared by all types, prompts and models, so raising ρ replaces one clean passage by its own twin and nothing else. A condition $\theta = (M, \pi, \tau, \rho)$ (model, prompt, type, ratio) yields for each question one outcome in $\{\textsc{Correct}, \textsc{Poisoned}, \textsc{Both}, \textsc{OtherWrong}, \textsc{Abstain}, \textsc{NoAnswer}\}$ and a conflict flag, set when the response says that the passages disagree or may be unreliable. Accuracy (Acc), the attack success rate (ASR, the share of Poisoned) and the abstention, missing-answer and flag rates are shares of all N responses, and all comparisons are paired.

2.3 Data and poisoned contexts

From the HotpotQA development set (Yang et al. 2018) (distractor setting) we keep questions whose answer is not *yes* or *no*, appears in exactly one supporting paragraph (the answer passage), and whose label is bridge (4,672 of 7,405), and draw with a fixed seed 50 pilot questions, used only to build and validate the scoring, and 500 disjoint main questions. Gemma-2-9B-it, a model we do not evaluate, wrote the paraphrases and generated the target answers, both checked automatically and by hand (details in the repository). The four poison types replace every occurrence of a^* by $\tilde{a}$; Swap changes nothing else, while Correct, Authority and Recency append one templated sentence claiming that other sources are wrong, that the fact is officially confirmed, or that the passage is the latest update. Passage order and the bridge position are random per question and recorded, since models use the start and end of a context more than the middle (Liu et al. 2024). A retrieval check runs BM25 (top five, seeded tie-breaking) over all clean, poisoned and bridge passages plus the distractor paragraphs, a poisoned-corpus setting (Zou et al. 2025). An audit of 73 questions with warning signs flagged 9 (1 wrong gold answer, 8 comparison questions past the bridge filter); all stay in the primary analysis.

2.4 Models, prompts and decoding

Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3 run in bfloat16 with vLLM on one A100 GPU. Each prompt holds the instruction, the numbered passages, the question and an output instruction to reason briefly and end with

"Answer: ⟨answer⟩" or "Answer: unknown". The plain prompt only asks to answer from the passages; the consistency prompt adds "First check whether the passages agree with each other, then answer."; the reliability prompt adds "First judge how reliable each passage is, then answer." Models are never told that passages may be false. Besides 48 poisoned and 3 clean conditions per model there are a closed-book and 12 retrieval conditions; greedy decoding gives the primary results, and five samples at temperature 0.7 measure self-consistency (512-token cap).

2.5 Scoring and statistical analysis

The answer after the last "Answer:" is matched to the gold and target answers by whole-word containment, and ordered rules assign the outcome (rules in the repository). A blind pilot audit found the CORRECT, POISONED and ABSTAIN rules exact but half of the OTHERWRONG answers to be aliases (Kamalloo et al. 2023), so rule-based accuracy is a lower bound; an LLM judge that failed a check fixed in advance gives an exploratory upper bound. Gemma-2-9B-it, reading the response alone, assigns the conflict flag ($\kappa = 0.735$ against 300 hand labels, 0.57 for Qwen); as it over-flags, flags are read against each model's $\rho = 0$ baseline. A hallucination is an OTHERWRONG answer with a name or number found in no passage (precision 9/13; 0 of 18 inventions missed). Self-consistency is the share of five samples agreeing with the most frequent answer.

Four confirmatory families, fixed before the main run, use Holm's correction within each ($\alpha = 0.05$): **F1** (H1), per model, the effect of ρ on POISONED (logistic GEE clustered by question); **F2** (H2), exact McNemar tests of each meta-prompt against the plain prompt, per model, on POISONED at $\rho > 0$ and on CORRECT and ABSTAIN at $\rho = 0$; **F3** (H3), per model, the odds ratio of POISONED for flagged against unflagged responses, adjusted for ρ, prompt and type; **F4** (H4), the interactions of ρ and prompt with type and model. Everything else is exploratory, with Wilson or question-clustered bootstrap intervals. $N = 500$ was kept although a pilot power check left a quarter of comparisons below 0.80 power for five points, so a non-significant cell is never read as no effect.

2.6 Revisions to the planned design

Five rules changed after pilot checks and before any main-run response (target generation, output cap, BM25 tie-breaking, two added outcomes, the judge as an upper bound), none touching the research questions, hypotheses, factors or tests; each is documented with its triggering check in the project notes.

3 Experimental Results

3.1 Experimental setup

All 192 conditions ran on one A100 GPU in 39.1 compute units, giving 576,000 completions (96,000 greedy, 480,000 sampled) and 188,047 judge labels, with no resumed run. Each request had its own seed; with one GPU type, residual hardware nondeterminism remains. The raw outputs had the same SHA-256 fingerprint on the cloud

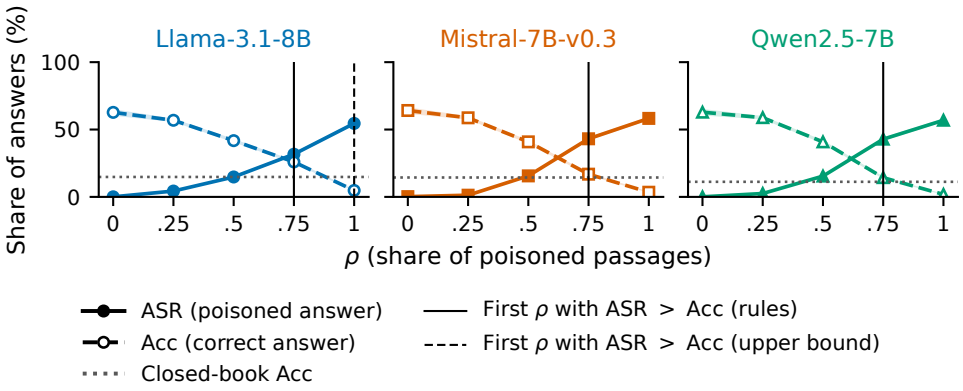


Fig. 1 Accuracy (open markers) and ASR (filled) against ρ per model, pooled over prompts and types, with 95% Wilson intervals. Dotted: closed-book accuracy; vertical lines: first ρ with ASR above accuracy (solid: rules; dashed: upper bound).

machine and locally, and were scored by the evaluator frozen at tag `evaluator-v1`. Estimates with intervals are in Tables T1–T3 of the repository.

3.2 Robustness to poison (RQ1)

Every model follows the passages (Fig. 1): from $\rho = 0$ to $\rho = 1$, ASR rises from 0.1–0.2 % to 54.5–58.3 % and accuracy falls from 62.7–64.1 % to 1.8–4.8 %. F1 is significant for every model after Holm's correction, with odds ratios of 50.3–64.9 between $\rho = 0$ and $\rho = 1$, large because almost no answer is poisoned at $\rho = 0$. The steepest rise (exploratory) lies between $\rho = 0.50$ and 0.75 for Mistral and Qwen (+27.5 and +27.4 points), when the poison becomes the majority, and between 0.75 and 1.00 for Llama (+22.9). H1 placed it where the poison stops being outnumbered, $\rho = 0.5$, which with four passages is a 2–2 tie: the rise holds, its location does not, and H1 is *partly supported*. Its last part holds: at $\rho = 1$ accuracy is below closed-book accuracy (11.2–14.8 %), and questions answered correctly without passages are still poisoned in 55.2–63.7 % of cases. Under the rules ASR first exceeds accuracy at $\rho = 0.75$ for every model; the judge-based upper bound adds 10.1–22.7 points to accuracy (at most 5.8 to ASR) and moves Llama's crossover to $\rho = 1.00$. Missing answer lines matter only for Llama (6.2 %, 5.1 points of them from the 512-token cap).

3.3 Effect of meta-prompts (RQ2)

Over all poisoned conditions the meta-prompts lower ASR in 5 of 6 confirmatory comparisons (F2), changing it by −1.8 to −7.7 points; for Qwen the reliability prompt shows no detectable effect (+0.2, 95% CI −1.2 to 1.7), which rules out large effects but not small ones. On clean contexts abstention rises in 5 of 6 comparisons (+4.2 to +14.0 points), and accuracy falls significantly only for Mistral (−4.2 points with the consistency prompt, −12.8 with the reliability prompt). The per-ρ differences (exploratory; figure in the repository) are not concentrated at intermediate ρ for

every model: for Mistral with the reliability prompt the reduction grows with ρ and is largest at $\rho = 1$, where no clean passage is left to compare against, and Llama with the reliability prompt changes only at $\rho = 1$. Of the answers poisoned under the plain prompt, 64.3–80.0 % stay poisoned under a meta-prompt; for Mistral with the reliability prompt the others are mostly abstentions (18.6 %, against 7.1 % correct). The prompts reduce ASR mostly by refusing, not by recovering the true answer, and H2 is *partly supported*. Single cells are noisy: the median minimum detectable difference is 4.9 points, and 43 of 96 cells have under 80% power for five points.

3.4 Conflict awareness (RQ3)

With ρ, prompt and type fixed, flagged responses are less often poisoned for all three models (F3, significant after Holm's correction; odds ratios 0.81–0.83), yet flagged answers are still poisoned in 21.2–26.0 % of cases: the test shows an association, not that noticing a conflict prevents poisoning. Flags are read against each model's clean baseline (2.1–30.8 %, partly inflated by the judge's over-flagging). Above it they peak at $\rho = 0.5$, where clean and poisoned passages are tied (+14.2 to +43.2 points), and fall at $\rho = 1$, where all passages agree (+1.5 to +16.9). H3 is *supported*, as an association.

3.5 Poison types, models and further measures (RQ4)

All 4 interaction blocks of F4 are significant (χ^2 from 22.2 to 71.4), but the differences behind them are small: pooled over $\rho > 0$ the poison types differ by at most 2.1 points within a model (3.7 at a single ρ), in no consistent order, while Llama is poisoned less often (26.3 %) than Mistral (29.6 %) and Qwen (29.4 %). H4 is *partly supported*: the interactions exist, but authoritative framing does not make a poison more effective here.

The remaining results are exploratory. At $\rho = 0.25$ Qwen is poisoned in 6.6 % of cases when the single poisoned passage comes first, against 0.8–1.2 % in the middle positions. With BM25, 50.2 % of the retrieved contexts contain no answer-bearing passage; elsewhere ASR follows the effective poisoned share. Hallucinations are rare with controlled passages (0.6–2.2 % of responses) and reach 5.3 % for Llama with retrieval. Self-consistency over five samples ranges from 63.7 to 91.0 %; it is highest for Qwen and with the plain prompt, and falls from $\rho = 0$ to $\rho = 1$ for every model and prompt. Without the 9 audit-flagged questions all 28 confirmatory conclusions hold; without the 70 questions whose poison leaks the gold answer 26 of 28 hold, as Llama's reliability-prompt ASR reduction and Mistral's consistency-prompt accuracy cost lose significance.

4 Concluding Remarks

The three open models we tested follow retrieved evidence almost mechanically. Their answers track the share of poisoned passages, attack success overtakes accuracy once the false passages are the majority, and full poisoning overrides answers the models give correctly without any passage. This matches the majority behaviour reported by Jin et al. (2024) and the overriding of correct priors reported by Wu et al. (2024),

here measured with paired contexts that change one passage at a time. The practical consequence is that a 7–8B model in a RAG system is no more reliable than the share of trustworthy passages its retriever returns.

Prompt-level defences help less than they appear to. The meta-prompts lower attack success by a few points, mostly by making the model abstain rather than recover the true answer, and the largest reduction comes with the largest accuracy cost on clean evidence (Mistral with the reliability prompt): the user gains caution, not discernment. Conflict statements are informative, since flagged answers are poisoned less often, but a flag is neither a guarantee nor shown to be a cause: flagged answers are still often poisoned, and the judge that detects flags over-flags. Poison framing barely mattered; the models differed more than the poison types.

What the study does not show matters as much. Accuracy is a bracket between the rule-based lower bound and the judge-based upper bound, and Llama's crossover depends on which bound is used. H1 placed the majority at $\rho = 0.5$, which with four passages is a tie; we kept the wording and report where the change actually occurs. The four poison types share one false answer and differ by one sentence, a weak manipulation of plausibility, and all poisons are synthetic. The flag result is an association measured by an imperfect judge ($\kappa = 0.57$ for Qwen). Many single cells are under-powered, so a non-significant cell is not evidence of no effect. HotpotQA contributed comparison questions with implausible targets and some answer leaks, although the sensitivity analyses left nearly all conclusions unchanged. The retrieval check uses a small poisoned BM25 corpus, and all results come from three 7–8B models on one dataset.

Future work should test larger models, natural rather than synthetic conflicts such as outdated or disputed facts, dense and hybrid retrievers over realistic corpora, and defences that act on the passages, such as source scoring or cross-passage verification, rather than on the prompt alone.

Use of AI tools. Claude Opus 5.5 (Anthropic) was used through a chat interface as an assistant to help write parts of the code, debug it, interpret the results, resolve technical issues and organise the LaTeX code of this report; all outputs were reviewed and revised by the author.

Code and data availability. Code, constructed data, analysis outputs, tables, figures and the generated result macros are in the public repository https://github.com/AngelaAldaoud/in-rag-we-trust, with instructions to reproduce every table and figure. The raw model outputs are not in the repository; they are available from the author on request, and their integrity can be checked against the SHA-256 fingerprint recorded in the project notes.

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